

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

TECHNICAL SKILLS

Business Analysis: Data Analysis, Business Analysis, Exploratory Data Analysis (EDA), Requirements Analysis, Process Analysis, Operational Analysis, Trend Analysis, Root-Cause Analysis, Decision Support, Predictive Analytics, Statistical Analysis, Performance Analysis

Data: SQL, Python, Pandas, NumPy, Data Cleaning, Data Validation, Data Transformation, Data Aggregation, SQL Server, PostgreSQL, MySQL, Oracle, ClickHouse, MongoDB, Joins, Subqueries, CTEs, Window Functions

Reporting: Data Visualization, Dashboard Development, KPI Monitoring, KPI Reporting, Performance Reporting, Analytical Reporting, Grafana, Streamlit, Data Storytelling

Process & Operations: Process Improvement, Process Optimization, Workflow Automation, Quality Analysis, Data Quality, Requirements Gathering, Documentation, Stakeholder Collaboration, Cross-Functional Collaboration

Tools & Methodologies: Git, GitHub, Jira, Docker, Linux, Agile, Scrum, SDLC, Technical Documentation

PROFESSIONAL EXPERIENCE

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

- Built data processing workflows to transform raw video and image data into structured datasets containing 4,569 frames, 11,422 bounding boxes, and 55 expert-annotated sequences
- Performed data cleaning, preprocessing, validation, segmentation, and normalization to improve data quality and consistency for downstream analysis
- Analyzed structured datasets to identify temporal patterns, evaluate performance, and support data-driven assessment
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID, achieving 91.83% keypoint mAP, 0.93 MOTA, and 0.91 IDF1
- Compared results across processing stages to identify performance trends, evaluate outcomes, and support process improvements

Project Engineer, Soliton Technologies, India

Jun 2023 – Jun 2024

- Served as sole developer on a Medtronic project, automating approximately 94% of manual test cases while meeting medical-device quality, validation, and compliance requirements
- Analyzed existing testing workflows and requirements to identify automation opportunities, reduce repetitive manual processes, and improve operational efficiency
- Designed and implemented a C# WPF application integrating Python and LabVIEW workflows to automate testing, collect results, and streamline user workflows
- Developed automated workflows for data collection, validation, reporting, and process monitoring, improving test coverage and consistency
- Analyzed test results and system behavior to identify failures, investigate data inconsistencies, and perform root-cause analysis
- Collaborated with cross-functional engineering teams to gather requirements, troubleshoot issues, optimize processes, document solutions, and implement process improvements

Intern, Soliton Technologies, India

Jul 2022 - May 2023

- Developed Python and LabVIEW-based automation workflows supporting verification, validation, data processing, and reporting
- Contributed to MAVIS, an object-oriented automation framework integrating Python and LabVIEW for scalable testing and data collection
- Assisted with workflow development, debugging, data validation, test-result analysis, and technical documentation

PROJECTS

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, SQL, Python, ClickHouse, Grafana

- Developed an end-to-end analytics platform to ingest, transform, aggregate, and analyze high-frequency medical IoT data
- Implemented rolling 5-second statistical analysis and Z-score anomaly detection to identify abnormal device behavior and monitor performance
- Designed analytical data structures optimized for high-volume queries and real-time reporting
- Built Grafana dashboards to monitor device-level KPIs, trends, performance metrics, and anomalies
- Processed streaming data with sub-second latency, enabling timely identification of abnormal device behavior

Expense Tracker with Predictive Analytics

C++, Python, MySQL, SQL

- Developed an expense analytics application to store, categorize, aggregate, and analyze historical transaction data
- Wrote SQL queries to generate category-level spending summaries, trends, and reports
- Applied Python-based analytics to identify spending patterns and generate next-month expense forecasts
- Integrated database and analytical workflows to transform raw transaction data into actionable spending insights